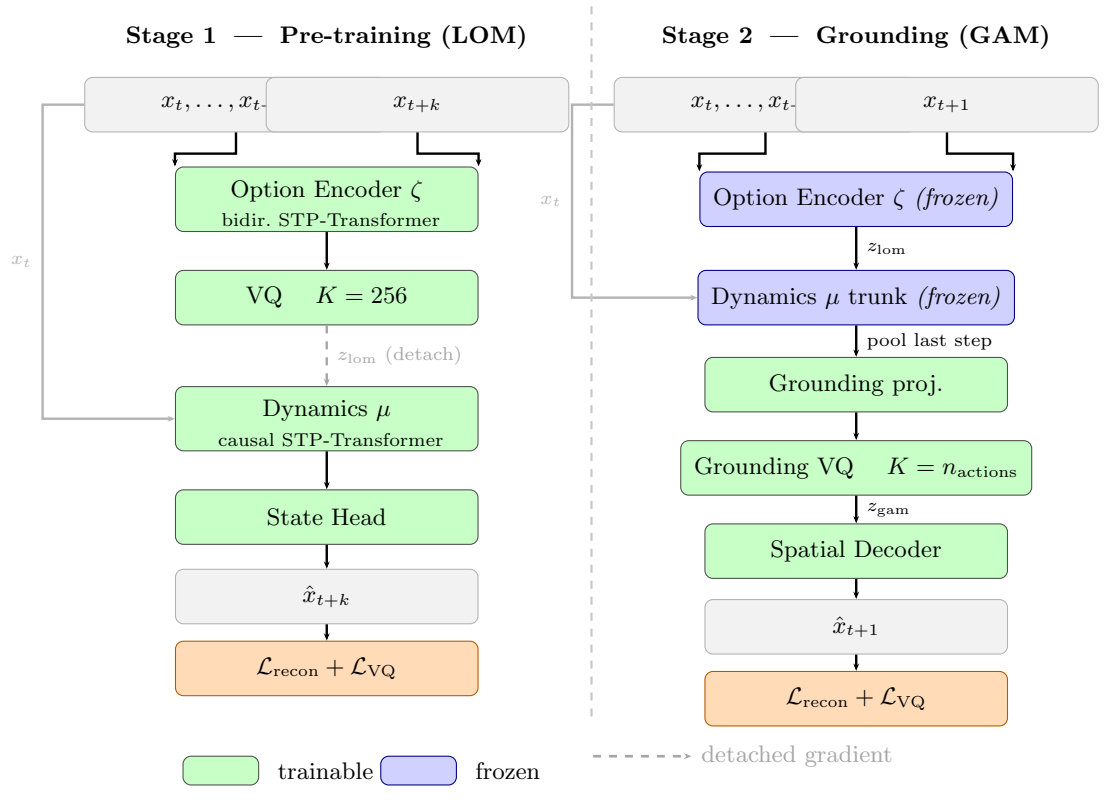

Stage 1 — Pre-training (LOM)
$x_t, \ldots, x_{t-}$
x_{t+k}
x_t
Option Encoder ζ
bidir. STP-Transformer
VQ $K = 256$
z_{lom} (detach)
Dynamics μ
causal STP-Transformer
State Head
$\hat{x}_{t+k}$
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{VQ}}$
Stage 2 — Grounding (GAM)
$x_t, \ldots, x_{t-}$
x_{t+1}
x_t
Option Encoder ζ (frozen)
z_{lom}
Dynamics μ trunk (frozen)
pool last step
Grounding proj.
Grounding VQ $K = n_{\text{actions}}$
z_{gam}
Spatial Decoder
$\hat{x}_{t+1}$
$\mathcal{L}_{\text{recon}} + \mathcal{L}_{\text{VQ}}$
trainable
frozen
detached gradient